

~~#include <stdio.h>~~
~~void greatNum (int~~

Q1 - Find the equations whose roots are

$$2 \cos \frac{2\pi}{7}, 2 \cos \frac{4\pi}{7}, 2 \cos \frac{6\pi}{7}$$

Q2 - Use De-Moivre's theorem to solve

$$x^7 + x^4 + x^3 + 1 = 0$$

Q3 - Find all values of $(1+i\sqrt{3})^{1/3} + (1-i\sqrt{3})^{1/3}$

Q4 - Separate real and imaginary parts

(i) $\exp(z^2)$, $\exp(5+3i)^2$

(ii) $\cos^{-1}(\cos \theta + i \sin \theta)$

(iii) $\log \cos(x+iy)$

Q5 - Find modulus and argument of

$$(1-i)^{1+i}, i \log(1+i)$$

Q6 - Find all roots of (i) $\cos z = 2$

(ii) $\tanh z + 2 = 0$

Q7 - Prove that $\tan \left[i \log \left(\frac{a-ib}{a+ib} \right) \right] = \frac{2ab}{a^2-b^2}$

Q8 - Prove that.

(i) $\sin^{-1}(ix) = 2n\pi + i \log(\sqrt{1+x^2} + x)$

(ii) $\sin^{-1}(\operatorname{cosec} \theta) = \frac{\pi}{2} + i \log \cot \theta$

(iii) If $\cos(\theta + i\phi) = \cos \alpha + i \sin^2 \alpha$

then show $\sin^2 \theta = \pm \sin \alpha$

$\cos 2\theta + \cosh 2\phi = 2$

Q9 - If $\sin(A+iB) = x+iy$,

prove, $\frac{x^2}{\cosh^2 B} + \frac{y^2}{\sinh^2 B} \geq 1$, $\frac{x^2}{\sin^2 A} - \frac{y^2}{\cos^2 A} = 1$

Q10 - From the definition of $\sin z$ and $\cos z$

prove (i) $\cos^2 z = 2 \cos^2 z - 1$

(ii) $\frac{\sin^2 z}{1 - \cos^2 z} = \cot z$

Q11 - If $z = e^{i\theta}$, show that $\frac{z^2-1}{z^2+1} = i \tan \theta$

Q12 - Show that $\tan^{-1} z = \frac{i}{2} \log \left(\frac{i+z}{i-z} \right)$